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INCH-POUND

MIL-M-38510/101G
AMENDMENT 4
19 FEBRUARY 1999
SUPERSEDING
MIL-M-38510/101G
AMENDMENT 3
17 NOVEMBER 1998

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, OPERATIONAL AMPLIFIERS, MONOLITHIC SILICON

Inactive for new design after 13 July 1995

This amendment forms a part of MIL-M-38510/101G, dated 8 May 1989, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGE 1

1/, line 2, delete “dice” and substitute “die”.

* 1.2.3 Case outlines. Add outline letter “Z”. Add case outline description; “See figure 1, (10-lead), flat pack”.

PAGE 2

* 1.5 Power and thermal characteristics. Add; package “10-lead FP”, case outline “Z”, maximum allowable power dissipation “330 mW at TA = +125◦C”, maximum θ_{JC} “21◦C/W”, maximum θ_{JA} “225◦C/W still air, 142◦C/W 500 LFPM”.

The attached insertable replacements pages listed below are replacements for stipulated pages. When the new pages have been entered in the document, insert the amendment as the cover sheet to the specification.

Replacement page	Page replaced
11a	new
11b	new

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PAGE 3

3.1, delete and substitute the following;

"3.1 Item requirements. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein. The individual item requirements for devices marked with the "Q" certification mark shall be in accordance with MIL-I-38535 and as specified herein or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

PAGE 4

Table I, add new footnote "1/" after the title which will state the following;

"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

Table I, input offset voltage temperature sensitivity, symbol column, delete "ΔV_{IO}" and substitute "$\Delta V_{IO}/\Delta T$". Also, footnote "1/" will be renumbered as footnote "2/".

Table I, input offset current temperature sensitivity, symbol column, delete "ΔI_{IO}" and substitute "$\Delta I_{IO}/\Delta T$". Also, footnote "1/" will be renumbered as footnote "2/".

Table I, input bias current test, footnote "1/" will be renumbered as footnote "2/". Device types 01, 02, 03, 05, 07, and 08, delete "+1 nA min" and substitute "-0.1 nA min".

PAGE 5

TABLE I, add new footnote "1/" after the title which will state the following;

"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

TABLE I, Adjustment for input offset voltage (positive) test, footnote "2/" will be renumbered as footnote "3/".

TABLE I, Adjustment for input offset voltage (negative) test, footnote "2/" will be renumbered as footnote "3/".

TABLE I, Output short-circuit current (for positive output) test, footnote "3/" will be renumbered as footnote "4/".

TABLE I, Output short-circuit current (for negative output) test, footnote "3/" will be renumbered as footnote "4/". Also, delete in its entirety and substitute the following:

Characteristics	Symbol	Conditions (Figure 4, unless otherwise specified		Limits								Units
				01,02,08		03,05		04,06		07		
				Min	Max	Min	Max	Min	Max	Min	Max	
Output short-circuit (for negative output)	I_{OS}-	$\pm V_{CC}$ = 15 V, T_A = -55°C to +125°C $t \leq 25$ ms 4/			+60		+60		+15			mA
			T_A = +25°C to+125°C								+65	mA
			T_A = -55°C								+80	mA

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PAGE 5

TABLE I, Supply current test, footnote "4/" will be renumbered as footnote "5/".

TABLE I, Both open loop voltage gain (single ended) tests, footnote "5/" will be renumbered as footnote "6/". Also, footnote "6/" will be renumbered as footnote "7/".

TABLE I, Transient response rise time tests, footnote "7/" will be renumbered as footnote "8/".

TABLE I, Transient response overshoot tests, footnote "7/" will be renumbered as footnote "8/".

TABLE I, Both slew rate tests, footnote "8/" will be renumbered as footnote "9/". Also, footnote "9/" will be renumbered as footnote "10/".

TABLE I, Slew rate test, SR(+), for device type 07, delete "+50" and substitute "+40". Also, delete "+40" and substitute "+30".

TABLE I, Slew rate test, SR(-), for device type 07, delete "+50" and substitute "+40". Also, delete "+40" and substitute "+30".

PAGE 6

TABLE I, Settling time test, footnote "10/" will be renumbered as footnote "11/". Also, limits column for device type 07, delete in its entirety and substitute the following:

Characteristics	Symbol	Conditions (Figure 4, unless otherwise specified	Limits								Units
			01,02,08		03,05		04,06		07		
			Min	Max	Min	Max	Min	Max	Min	Max	
Settling time 11/	t_S+	See figure 5, T_A = +25°C								800	ns
		T_A = -55°C to +125°C								1200	ns
	t_S-	See figure 5, T_A = +25°C								800	ns
		T_A = -55°C to +125°C								1200	ns

Add a new footnote "1/" which will state;
"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

Footnote "1/" will be renumbered as footnote "2/".

Footnote "2/" will be renumbered as footnote "3/".

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PAGE 6

Footnote “3/” will be renumbered as footnote “4/”.

Footnote “4/” will be renumbered as footnote “5/”.

Footnote “5/” will be renumbered as footnote “6/”.

Footnote “6/” will be renumbered as footnote “7/”.

Footnote “7/” will be renumbered as footnote “8/”.

Footnote “8/” will be renumbered as footnote “9/”.

Footnote “9/” will be renumbered as footnote “10/”.

Footnote “10/” will be renumbered as footnote “11/”.

PAGE 7

4.1, delete and substitute the following;
"4.1 Sampling and inspection. Sampling inspection procedures shall be in accordance with MIL-M-38510 and methods 5004, 5005, and 5007, as applicable, of MIL-STD-883, except as modified herein. For devices marked with the "Q" certification mark, sampling and inspection procedures shall be in accordance with MIL-I-38535 or as modified in the device manufacturer's Quality Management (QM) plan. The modification in the QM plan shall not effect the form, fit, or function as described herein."

PAGE 8

4.4.2b, delete the last sentence and substitute the following; “If alternate burn-in conditions are used, the circuit shown in either Figure 7 or Figure 8 shall be used.”

4.2.d, line 3, delete “either”.

4.4.3.c, lines 3 and 4, delete and substitute the following:
“The sample size series for subgroup 12 shall be 5, and 13 shall be 7 for class B devices (see MIL-PRF-38535, Appendix D).”

PAGES 9 THROUGH 15

* FIGURE 1. Delete the title “Terminal connections” and substitute “Case outline and terminal connections”.

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PAGE 18

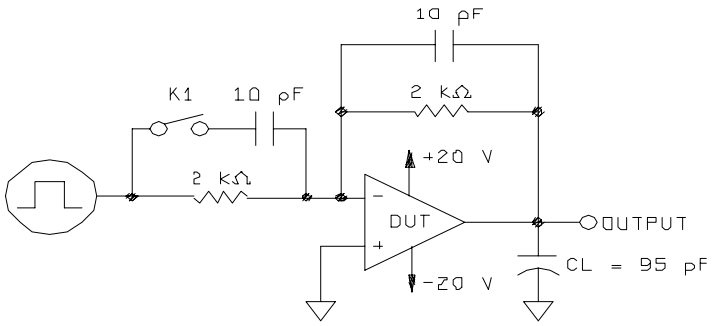
FIGURE 4 at the center upper half of the circuit at I_{OS} meter, delete "S_7" and substitute "S_6". Also, at the center bottom of the circuit S_{4A} position 7, delete the "200 Ω" resistor and substitute "200 kΩ".

PAGE 21

6/, delete and substitute the following; "6/ Device type 01, all case types, and device type 02, case A, B, C, and D only."

PAGE 23

FIGURE 5, Add the following diagram as an alternate test circuit for device type 07 only.



NOTES:

1. K1 is closed for small signal tests (Tr and P.O.) and is open for large signal tests (±slew rate, ±Ts).
2. Input signal is a -50 mV to 0 mV pulse train for small signal tests and -5 V to +5 V pulse train for large signal tests.
3. Tr of the input signal is < 10 ns for the small signal tests.

PAGE 24

FIGURE 5, add note: "9. C_L capacitance specified includes stray, jig, and probe capacitance."

PAGE 27

FIGURE 8. Note 2, delete and substitute the following; "For device types 01 and 04, V_{IN} = 3.0 V dc. For device type 03, V_{IN} = +3 V or -3 V."

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PAGE 30

TABLE III, test numbers 9-16, device types 01, 02, 03, 05, 07, and 08, delete “+1 nA min” and substitute “-0.1 nA min”.

PAGE 31

TABLE III, test numbers 35-42, device types 01, 02, 03, 05, 07, and 08, delete “+1 nA min” and substitute “-0.1 nA min”.

PAGE 32

TABLE III, test numbers 61-68, device types 01, 02, 03, 05, 07, and 08, delete “+1 nA min” and substitute “-0.1 nA min”.

PAGE 33

Table III, test numbers 81 and 83, conditions column, delete “-15 V” and substitute “+15 V”.

Table III, test numbers 82 and 84, conditions column, delete “+15 V” and substitute “-15 V”.

Table III, test number 86, conditions column, delete “2 kΩ” and substitute “10 kΩ”.

PAGE 34

Table III, test numbers 91, 93, 101, and 103, conditions column, delete “-15 V” and substitute “+15 V”.

Table III, test numbers 92, 94, 102, and 104, conditions column, delete “+15 V” and substitute “-15 V”.

Table III, test numbers 96 and 106, conditions column, delete “2 kΩ” and substitute “10 kΩ”.

Table III, test numbers 109 and 110, device type 07, delete “50” and substitute “40”.

Table III, test number 110, conditions column, delete “V_{IN} = -5 V to +5 V” and substitute “V_{IN} = +5 V to -5 V”.

PAGE 35

Table III, test numbers 116 and 117, device type 07, delete “40” and substitute “30”.

Table III, test numbers 120 and 121, device type 07, delete “50” and substitute “40”.

TABLE III, add a sentence to footnote "1/" which will state the following;

"1/ For devices marked with the "Q" certification mark, the parameters listed herein maybe guaranteed if not tested to the limits specified herein in accordance with the manufacturer's QM plan."

3/, delete and substitute the following; “V_{IN}(adj) is not performed on device type 02, case I only, or on device types 04, 06, and 08, all case types.”

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NOTE: The margins of this amendment are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
NASA - NA

Preparing activity:
DLA - CC

Review activities:
Army - AR, MI, SM
Air Force - 19, 85, 99
Navy - AS, CG, MC, OS, SH

(Project 5962-1830)

Civil Agency Coordinating Activity:
DOT-FAA(RD-650)

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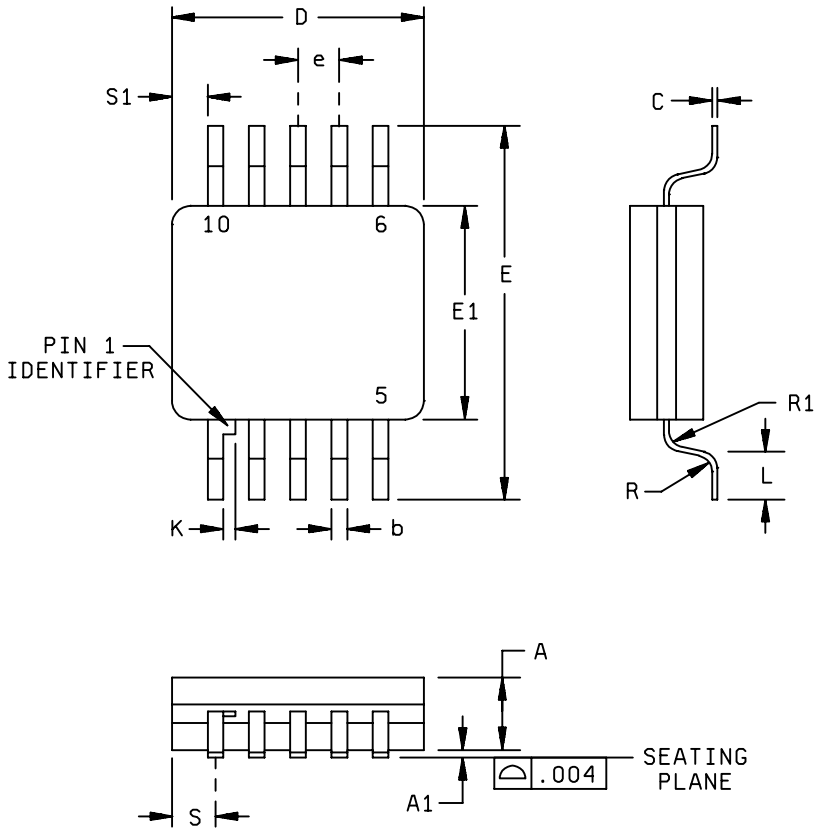
* FIGURE 1, add the following case outline

Case outline Z



NOTE:
Pin outs are the same as case H for device type 04. Pin 1 is NC, pin 2 is GUARD, pin 3 is -INPUT, pin 4 is +INPUT, pin 5 is GUARD, pin 6 is $-V_{CC}$, pin 7 is OUTPUT, pin 8 is $+V_{CC}$, pin 9 is OUTPUT COMP, and pin 10 is INPUT COMP.

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* FIGURE 1, add the following case outline

Case outline Z

Letter	Inches		Millimeters		Notes
	Min	Max	Min	Max	
A	.050	.080	1.27	2.03	
A1	.004	.012	0.10	0.30	
b	.015	.019	0.38	0.48	2
C	.004	.008	0.10	0.20	2
D		.270		6.86	
E	.400	.420	10.16	10.67	
E1	.236	.261	5.99	6.63	
e	.048	.052	1.22	1.32	
K	.008	.012	0.20	0.30	
L	.037	.043	0.94	1.09	
R	.013	.017	0.33	0.43	
R1	.013	.017	0.33	0.43	
S		.045		1.14	
S1	.005		0.13		

NOTES:
1. The U.S. government preferred system of measurement is the metric SI system. However, since this item was originally designed using inch pound units of measurement, in the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.
2. Maximum limit may be increased by .003 inches after lead finish is applied.

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